# COMP4135 SEM Weekly Report 1

**Group 2**

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# 2 1 Introduction

This weekly report is the very first one for this Unity game project. It mainly clarifies the division of people resource among project members and the project direction.

## 1.1 Description

This project is a group coursework of the Software Engineering Management module, which aims to design, develop, and test a game based on Unity. The core assignment focuses on documenting and collaborating. Our initial game direction candidate includes board chess game, complex card game (War of the Three Kingdoms). And we finally determined the direction of our game, Tactical Cards. It combines the grid based movement of board chess game with using cards to control from the card game, resulting a turn based strategy game. Generally, we use cards to control the pieces on the chessboard.

## 1.2 Motivation and Aims

Our boardcard game incorporates a variety of mechanisms and is a strategy game involving complex decision making processes. For teenagers, making multidimensional decisions (moving, attacking, defending, strengthening) under limited resources (action points) could be a beneficial exercise for their computational and planning abilities.

# 3 2 Requirements Engineering

## 2.1 Requirement Elicitation

End User:

The core target users, 12-20 in ages, casual strategy game players.

They need a game with easy to hand on and become challenging while time.

They need Clear fighting feedback.

Developer:

Responsible for full cycle design, development, test, and documentation.

## 2.2 Functional Requirements

### 2.2.1 Gaming Process

1 Player turn start: The system start player's turn, display "Your Turn" animation. Fully charged the action point (for example 3/3), draw 5 cards form the draw pile into player's hand. Display enemies' intentions.
2 During the player turn: The player reviews the cards in hand, select the cards and the targets (could be enemy or selected moving position). Play a card and consume the corresponding action points. Then the cards take effect. The used cards go into the discard pile.
3 Turn end: The cards in hand that have not been played during the turn are discarded into the discard pile, and the action points are cleared. Prepare to enter the opponent's turn.
4 Enemy turn: Enemy units execute their intentions.
5 State check: If there are still surviving units on the chessboard for both sides, return to 1. Otherwise, the game ends.

### 2.2.2 Gaming requirements

1 Some 6X6 grid battle boards, with different barrier locations.
2 Interface includes draw pile, hand zone, discard pile, play board, and some functional bottoms.
3 Different command cards including moving, strike, defense, strengthening.
4 Numerical calculation system, including health points, action points, attack and defense points, etc.
5 Display and execution of the intentions of enemy bot units.
6 The system should provide clear visual and animation feedback for all actions.

### 2.2.3 Product Requirements

1 The game should run stably on the Windows system with the frame rate remaining steadily above 30 FPS.
2 The system shall respond to all player inputs within 100ms, with no noticeable lag or delay.

3 The game interface shall be clean, intuitive, and suitable for teenage users, with all critical combat information (unit health, remaining card, turn state) clearly visible at all times.
4 The game shall have a maximum installation package size of no more than 1GB.
5 A well written game tutorial.

### 2.2.4 Organizational Requirements

4 All project codes shall be developed and managed using the team's GitLab/GitHub repository, following the standardized Git workflow and branch management rules defined by the team.
5 All project documents and deliverables shall follow the module's submission requirements and the team's unified document format standard, with traceable version records.

## 2.3 System Model

The following three UML diagrams illustrate the game architecture, state mechanism, and gameplay loop.

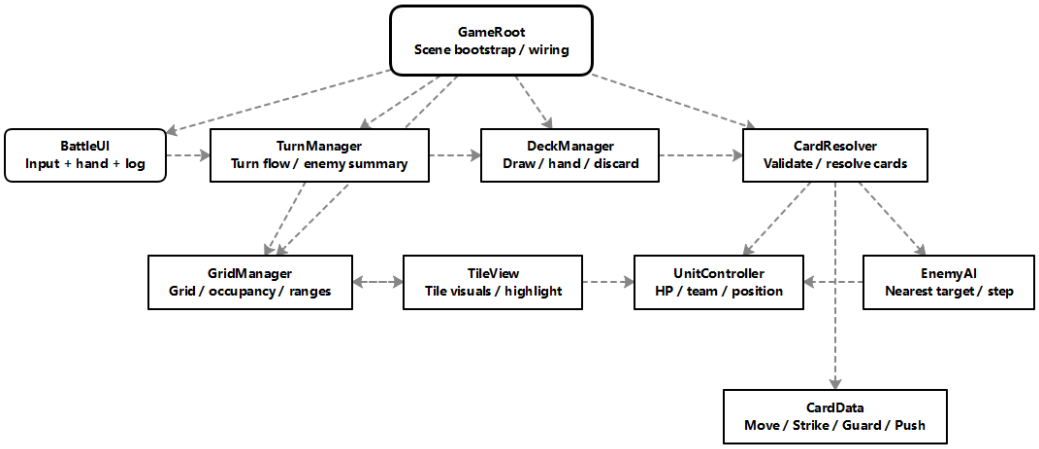



*Figure 1. Tactical cards demo architecture*

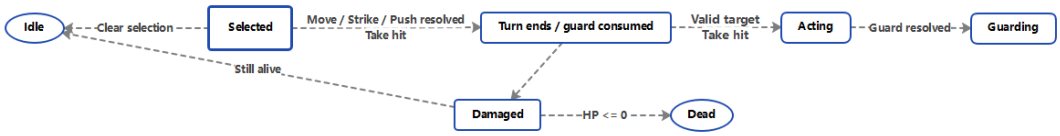



*Figure 2. Unit state machine*

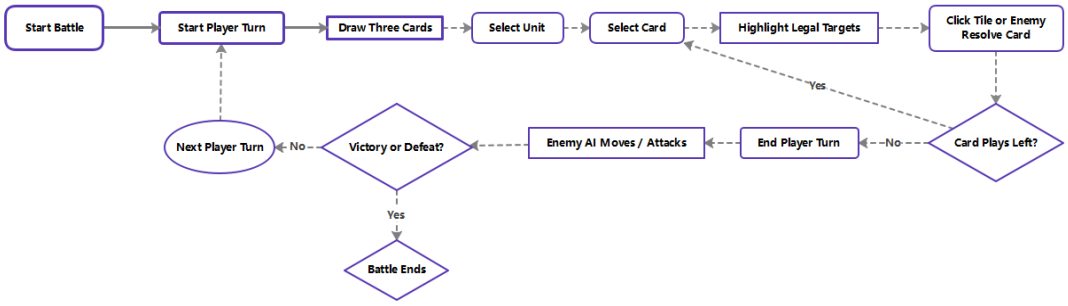



*Figure 3. Gameplay loop*

# 3 Summary and Reflections

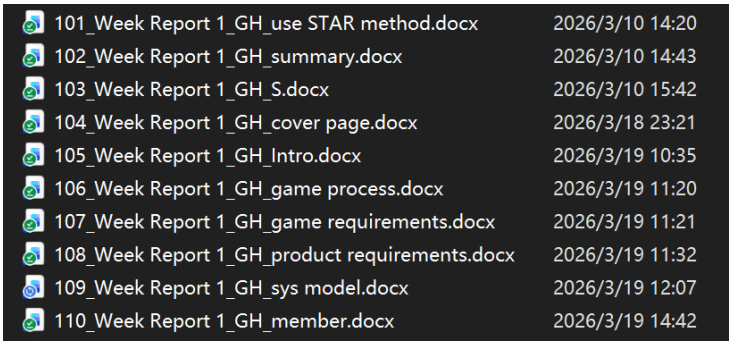
## 3.1 Team Members and the Division of Labor

- Guangjun HU: Requirements analysis, documentation
- Hongshuo HU:Modeling, Coding
- Cheng YE: Coding, documentation
- Zehan CHEN: Testing, Web development

## 3.2 Completed Process

Set up a GitHub repository [https://github.com/hu-zlatan/COMP4135-Group2-Project.git](https://github.com/hu-zlatan/COMP4135-Group2-Project.git)

Specified the naming standards for file management as shown in Figure 4; (Version Number)_(Name of the file)_(Editor)_(Comment)



| | |
|---|---|
| 101_Week Report 1_GH_use STAR method.docx | 2026/3/10 14:20 |
| 102_Week Report 1_GH_summary.docx | 2026/3/10 14:43 |
| 103_Week Report 1_GH_S.docx | 2026/3/10 15:42 |
| 104_Week Report 1_GH_cover page.docx | 2026/3/18 23:21 |
| 105_Week Report 1_GH_Intro.docx | 2026/3/19 10:35 |
| 106_Week Report 1_GH_game process.docx | 2026/3/19 11:20 |
| 107_Week Report 1_GH_game requirements.docx | 2026/3/19 11:21 |
| 108_Week Report 1_GH_product requirements.docx | 2026/3/19 11:32 |
| 109_Week Report 1_GH_sys model.docx | 2026/3/19 12:07 |
| 110_Week Report 1_GH_member.docx | 2026/3/19 14:42 |

*Figure 4. File management sample*

Assigned approximate role and tasks to each person.

Determined the basic game format.

## 3.3 Future Plan

Gradually carry out the development of the demo and form a game development requirements document. Manage the Git repository, conduct development on the develop branch, and merge tested versions into the main branch. Record bugs and form a bug library file. Maintain the meeting frequency to align ideas and requirements.

# Appendix A

## A.1 Meeting 1

Time

3.17

Attendance

- Guangjun HU
- Hongshuo HU
- Zehan CHE

Context

Understand everyone's game ideas and interests, determine the division of labor, and define the game direction. Assign tasks and set the deadlines and delivery requirements for individual tasks.

## A.2 Meeting 2

Time

3.18

Attendance

- Guangjun HU
- Cheng YE

Context

Introduce the current situation, align the requirements, assign tasks and set the delivery time. To standardize the project communication and ensure all meeting information is traceable, the team has defined a unified meeting minutes format to be used for all subsequent project meetings. The format includes the following core sections:

1. **Basic Meeting Info**: Time, location (online/offline platform), attendance (present/absent/leave), recorder
2. **Agenda**: Pre-determined discussion topics and priority order
3. **Key Discussion Points**: Detailed notes of core content discussed for each agenda item, including different opinions, decision-making processes and final conclusions
4. **Action Items**: Assigned tasks, responsible person, expected delivery time, delivery standard and follow-up reviewer
5. **Open Issues**: Unresolved problems, pending discussion topics and required preparation work for the next meeting
6. **Next Meeting Arrangement**: Tentative time, main agenda and pre-work requirements for all members